SDG Lens: Explainable Multi-Label Classification of Sustainable Development Goals

AI for Global Challenges

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Abstract

SDG Lens classifies text from the SDGi corpus into the 17 United Nations Sustainable Development Goals. The marker-facing pipeline trains a MiniLM-based multi-label neural model and a TF–IDF baseline over three random seeds and two training-set sizes, evaluates all artifacts on a fixed held-out test sample, and reports mean and standard deviation across seeds.

1 Pipeline

The full reproducible command is:

```
python main.py sweep
```

The pipeline writes trained artifacts to `artifacts/`, evaluation results to `results/`, manuscript-ready charts and tables to `outputs/`, and this compiled PDF to `manuscript/`.

2 Evaluation Results

Table 1: Mean ± standard deviation over seeds 42, 43, and 44.

Model	Train Size	Micro-F1	Macro-F1	Subset Acc.	Train Time
BERT	1,000	0.657 ± 0.011	0.654 ± 0.011	0.635 ± 0.034	11.5 ± 0.35
BERT	2,000	0.687 ± 0.010	0.682 ± 0.009	0.650 ± 0.024	22.0 ± 0.17
BERT	4,000	0.717 ± 0.003	0.706 ± 0.002	0.670 ± 0.013	44.6 ± 0.60
TF-IDF	1,000	0.611 ± 0.011	0.606 ± 0.014	0.452 ± 0.023	1.5 ± 0.06
TF-IDF	2,000	0.672 ± 0.012	0.667 ± 0.016	0.567 ± 0.004	2.4 ± 0.05
TF-IDF	4,000	0.700 ± 0.005	0.696 ± 0.009	0.595 ± 0.008	4.3 ± 0.08

3 BERT Model Analysis

4 Interpretability Note

The neural model records attention-based token examples as a proxy explanation. These should be interpreted as inspection aids rather than validated human rationales because

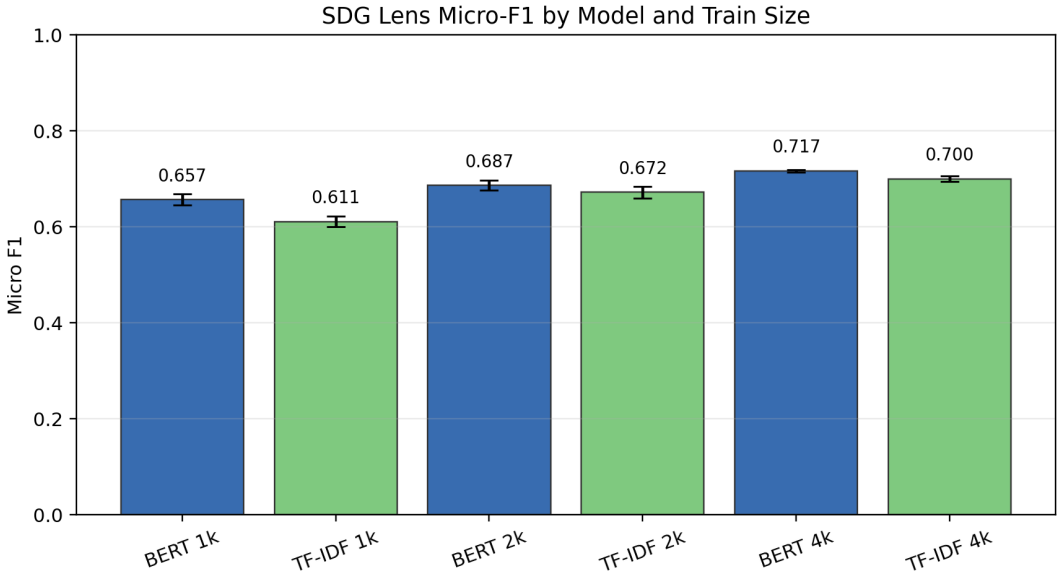

Figure 1: Micro-F1 by model type and training size. Error bars show standard deviation over seeds.

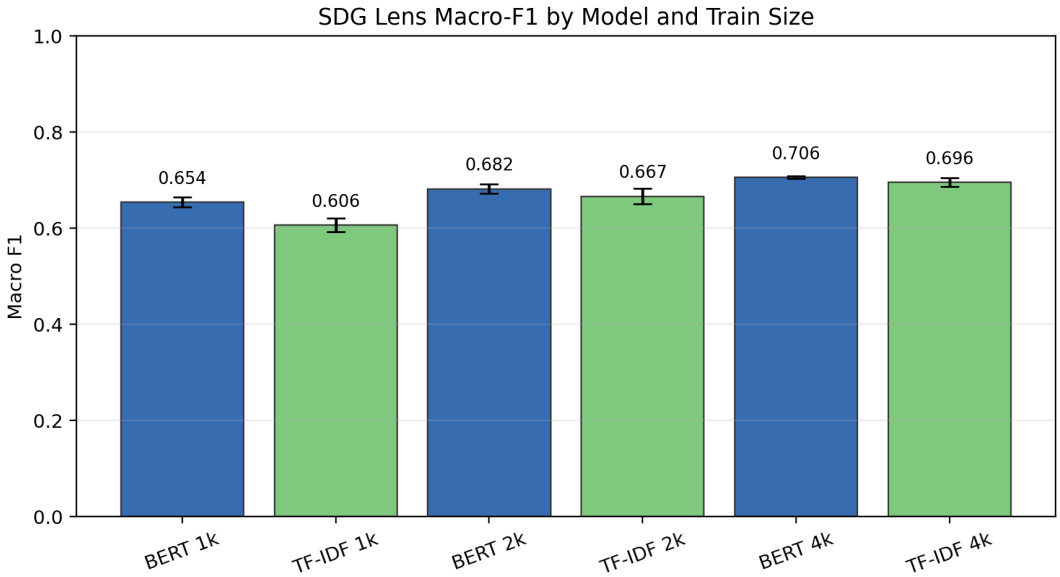

Figure 2: Macro-F1 by model type and training size. Error bars show standard deviation over seeds.

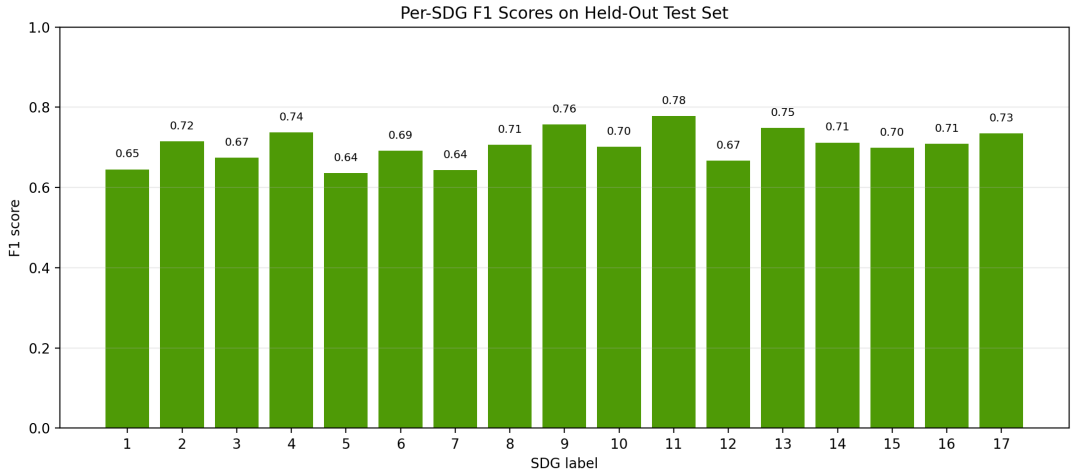

Figure 3: Per-SDG F1 scores for the BERT model. Green ≥ 0.6, amber 0.3–0.6, red < 0.3.

the SDGi corpus provides document-level SDG labels, not token-level rationale annotations.

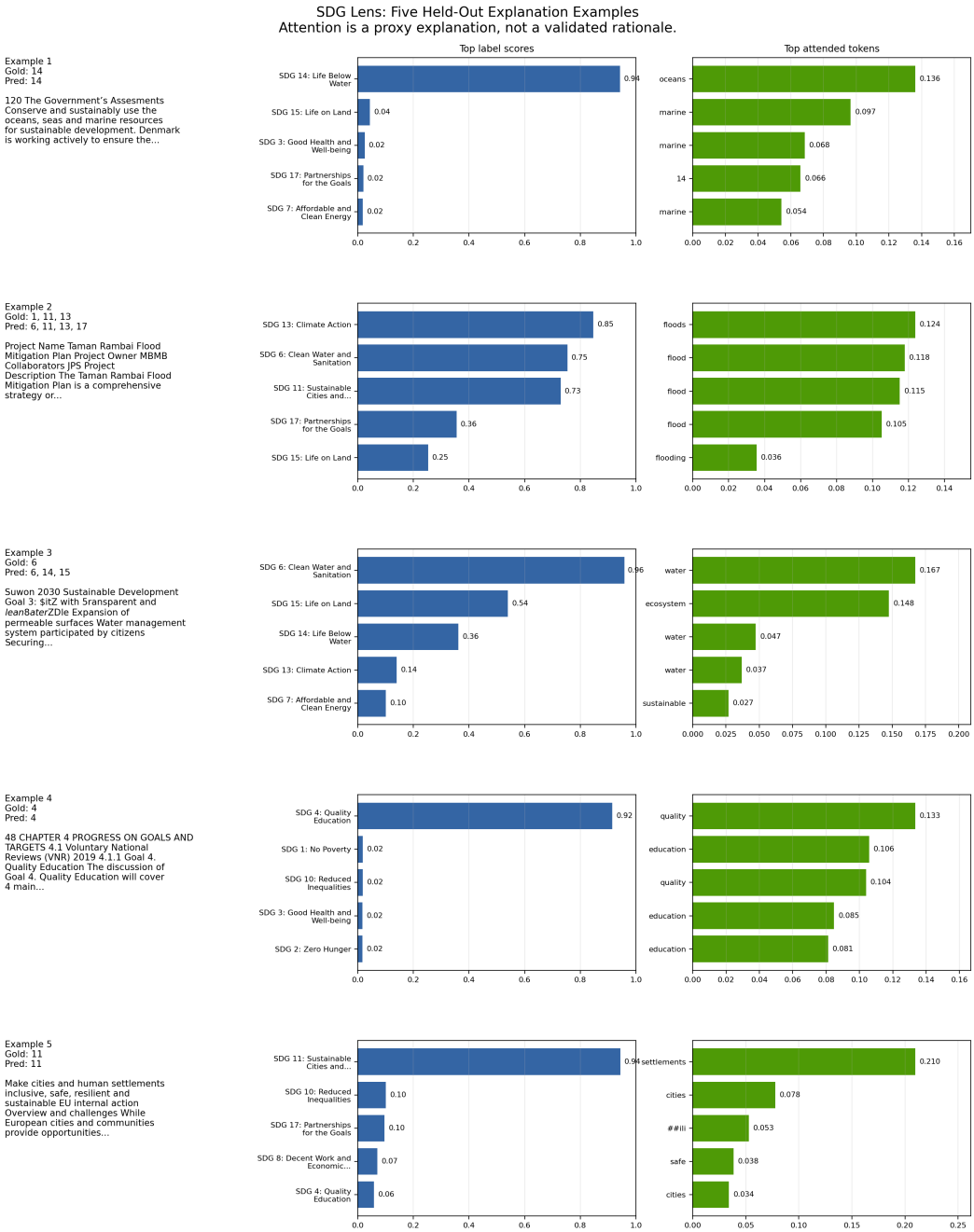

Figure 4: Top attended tokens and label scores for held-out explanation examples. Attention is a proxy explanation, not a validated rationale.